

CHAPTER 12

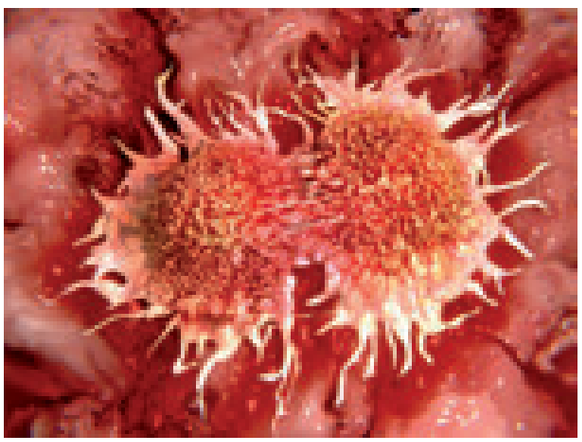
NUTRITION IN CANCER

OBJECTIVES

This chapter will enable students to

- Know what cancer is
- Know the causes, symptoms and major types of cancer
- Learn treatment procedures
- Understand role of specific nutrients in the prevention of cancer
- Understand nutritional intervention in cancer

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Cancer is an uncontrolled division of abnormal cells in the body which may invade other normal and healthy tissues. Cancer is also referred to as a **"malignant tumor."** It can affect any part of the body, and originates from a single cell that transforms into a cancer cell through interactions between DNA/genes and external agents. Presently in India, it is a major cause of morbidity and mortality.

12.1 What is cancer?

The process of origin and development of cancer is termed as **"carcinogenesis."** It is a multistage process that consists of three phases: **Initiation, promotion** and **tumor progression.**

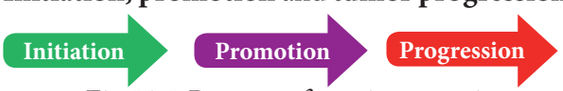


Fig 12.1 Process of carcinogenesis

Initiation:

Initiation involves a genetic change in the normal cells caused by their interactions with chemicals, radiation or viruses. These substances are termed as **carcinogens**. A carcinogen may be any substance that has the ability to initiate cancer. Carcinogens enter the body through inhalation (e.g., of cigarette smoke), ingestion of heavy metals and pesticides, and by the absorption of chemicals through skin. Initiation occurs rapidly but the initiated cells remain inactive until it is activated by a promoting agent.

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